

CS201 Cyber for Future Policymakers

Cyber for Future Policymakers, fall 2024

CS 12, CS 201, DHP D291

Syllabus v1.0

Course Description

Computer technology runs modern societies, underlying virtually every activity we do, from how the foods we eat arrive on our table to how we receive and consume information. Because cyber technology has such a profound influence on society, policymakers frequently face decisions on technical issues. Cyber for Future Policymakers is designed to provide students interested in policy, political science, and international relations aspects of cyber technology with an understanding of how these technologies work and the underlying issues of the policy debate. This course will cover Internet architecture and basic networking, the Web,

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Close and accept

Upon completion of this course, students will be able to:

- explain cyber technologies of current policy interest, both verbally and in writing
- discuss technical topics with people who lack a technical background
- write briefing papers on current and new technologies
- deliver presentations on current and new technologies
- CS 201 / DHP D291 students will also be able to discuss policy decision-making relevant to cyber technologies.

Course Topics

- Communications Networks
- Internet Architecture
- Understanding the World Wide Web (WWW)
- Cloud Computing
- Intellectual Property and Open Source Software (OSS)
- Cybersecurity and Privacy (e.g., cryptography, malware, cyberattacks, protection measures, etc.)
- Identity Management
- Artificial Intelligence (AI) and Machine Learning (ML)
- Quantum Computing

Prerequisites

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teach not only programming, but also introduce algorithmic thinking and data structures.

This requirement is a prerequisite; the programming course may not be taken simultaneously.

Which section should I register for?

- If you are an undergraduate student, register for CS 12. Undergraduates are not permitted to take the graduate-level sections.
- If you are a student in the Fletcher School, register for DHP D291. DHP P201 and CS 201 will also study the U.S. government structure related to cyber issues.
- If you are a graduate student in Computer Science, or are otherwise not in Fletcher, register for CS 201.
- Students in the CSPP program may register for either CS 201 or DHP D291.

Course Materials

- Required textbook:
 - Kernighan, Brian W. ***Understanding the Digital World: What You Need to Know about Computers, the Internet, Privacy, and Security***. Second Edition. Princeton University Press, 2017.
 - *The **second edition** is required.* There is new content that will be assigned reading, which is not present in the first edition.
- Laptop with the following software installed:

At least two web browsers (e.g., Firefox, Google Chrome)

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- A command line terminal software (built-in on Mac and Linux laptops; for Windows 10, use the Linux subsystem, use Ubuntu as distribution).

Class Times & Location

This class will be held *in-person*. Note that the course follows the **AS&E schedule**, not the Fletcher schedule for classes.

- **Mondays & Wednesdays, 10:30 – 11:45 am** (all sections), in **Joyce Cummings Center, room 140**
- **Fridays, 10:30 – 11:45 am** (CS 201 and DHP D291 sections only), on **Zoom** (link will be posted in Canvas)

The first class will be Wednesday, Sept. 4, 2024. The last day of class will be Monday, Dec. 9, 2024. Final presentation (in lieu of an exam) will be held during the class's final exam period, which is **Tuesday, Dec. 17, 2024, at 3:30-5:30 PM**.

Instructors & TAs

Dave Lillethun is the head instructor and will teach the Monday & Wednesday classes. Laurin Weissinger will teach the Friday classes.

Office hours and contact information will be posted in Canvas.

Assessments

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Assessment

• The first class will be Wednesday, Sept. 4, 2024. The last day of class will be Monday, Dec. 9, 2024. Final presentation (in lieu of an exam) will be held during the class's final exam period, which is **Tuesday, Dec. 17, 2024, at 3:30-5:30 PM**.

Labs that do not Meet Expectations may be resubmitted using the resubmission policy below.

Homeworks

There will be three written homework assignments to complete at home. These will be graded as Meets Expectations / Needs Improvements / or Not Assesable. Homeworks that do not Meet Expectations may be resubmitted using the resubmission policy below.

Briefing Papers

There will be three short briefing papers to write at home. These will be written based on a prompt that the instructors provide. They will be graded as Meets Expectations / Needs Improvements / or Not Assesable. Briefing papers that do not Meet Expectations may be resubmitted using the resubmission policy below. CS 201 / DHP D291 students will write all three briefing papers, whereas just two of the briefing papers will be assigned to CS 12 students.

Final Project

Paper

There will be one “project” briefing paper to write at home. (So-called because it counts towards the “Projects” category of your grade, not the same category as the other papers.) This will be written on a topic of your choice (subject to approval by the instructors). The length and expectations will be more for CS 201 / DHP D291 students, whereas CS 12

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Presentation

There will be one presentation on the same topic as your project paper. The length and expectations will be more for CS 201 / DHP D291 students than CS 12 students. These will be graded as Exceeds Expectations / Meets Expectations / Needs Improvements / or Not Assesable. Although the presentation cannot be resubmitted, you will have feedback on your project paper before you have to give the presentation, and you will have the opportunity to submit your slides for feedback in advance. So you will be able to know how good your content and slides are in advance, and make improvements before the presentation if necessary.

Threat Modeling Activity (Friday class)

CS 201 / DHP D291 students will engage in a an additional activity in a Friday class meeting, which will be “Risk and threat modeling in a policy context”. These will be graded as Exceeds Expectations / Meets Expectations / Needs Improvements / or Not Assesable. CS 12 students are not required to participate.

Late Work Policy

In this class, each students starts the semester with **10 “late tokens”** that are each worth an extra day (24 hours) on an assignment – lab, homework, or paper (including briefing papers and the project paper). There are no penalties for lateness as long as you use a sufficient number of late tokens. If you will not have work completed by the deadline, then email the instructor (contact information is in Canvas) *before* the deadline and tell 'm how many extra days you need. A corresponding number of late tokens

Assignment

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Sending the email is *required* to get the extension. (However, if there is an emergency situation that makes it impossible or unreasonable to contact us before the deadline, then just send the email as soon after the deadline as is reasonable.)

Note that the presentation and threat modeling activity are scheduled to occur in specific class periods, and therefore *the deadline **cannot** be extended*. However, make up assignments can be provided if you miss it due to a legitimate excused absence (such as illness). (Note that the presentations occur during the final exam period, so a make up presentation may require taking an “incomplete” grade in the class until the presentation can be rescheduled.)

Resubmission Policy

Labs, homeworks, and papers (including the briefing papers and the final project paper) that do not Meet Expectations may be revised and resubmitted. Only the grade from your last submission of an assignment counts towards your course grade. There are some restrictions on resubmissions, as follows:

- **We will only grade one resubmission per student per week.** If you resubmit more than one assignment at a time, we will grade one that week, the next one the next week, and so on. If you resubmit more assignments than there are weeks remaining in the semester, then we will only grade as many of them as there are weeks left.
 - i.e., Resubmit regularly through the semester; do *not* wait until the last few weeks to try and resubmit everything!

- You may not resubmit until you have received feedback on your

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enough before you even see our feedback, then use the Late Work Policy above. Deliberately submitting junk or incomplete work to try and use the resubmission policy as a replacement for the late policy will be considered an abuse of the systems.)

- You start the semester with **eight (8) resubmission attempts**. Each time you resubmit an assignment, you spend one (1) of your attempts to do so. Resubmitting an assignment multiple times will require an attempt for each resubmission. (For example, if you resubmit a briefing paper, the resubmission still doesn't Meet Expectations, and you revise and resubmit again, then that would cost you two (2) attempts – one for each resubmission.)

Course Grades

Your letter grade will be determined by how many assignments in each category are graded at a Meets Expectations or higher level. The scale is slightly different for CS 201 / DHP D291 students than for CS 12 students (partly due to the differences in the assignments). The “Projects” category consists of: the project paper, the presentation, and (for CS 201 / DHP D291 student) the threat modeling activity.

The way to read these is, using the CS 201 / DHP D291 grading table for example, to earn an A you need to Meet Expectations on all 4 labs, all 3 homeworks, all 3 briefing papers, the project paper, the presentation, and the threat modeling activity. To get a C, you need to Meet Expectations on 2 of the labs, 1 of the homeworks, 1 of the briefing papers, and 2 of the projects (project paper, presentation, or threat modeling activity). Also, doing better in some categories does not make up for under-performing in

Assignment

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	D	C	B	A
Labs (4)	1	2	3 _*	4
Homeworks (3)	1	1	2 _*	3
Briefing Papers (3)	1	1	2 _*	3
Final Project (3) (includes paper, presentation, and threat modeling activity)	1	2	3	3

* Indicates that all items in this category must still have a reasonably attempted first submission, even though that submission may not Meet Expectations. In other words, you cannot simply skip or not attempt one of the assignments. (Instructors reserve sole judgement over whether a submission demonstrates a reasonable attempt at the assignment.)

CS 12

	D	C	B	A
Labs (4)	1	2	3	4

Assessment

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Final Project (2) (includes paper and presentation)	1	1 **	2 **	2
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** Indicates that the grade for one briefing paper may be substituted for the project paper. So if the project paper does not meet expectations, but one more of the briefing papers than is needed for this grade meets expectations, that grade may be used on the briefing paper instead. (e.g., you can get a B if your Presentation meets expectations, and your project paper does not, but both briefing papers do meet expectations)

+ / -

Plus (+) and minus (-) on your letter grade will be determined by the number of “Exceeds Expectations” grades you receive on the Projects (project paper, presentation, and debate). If you get at least two (2) “Exceeds Expectations” grades, then you will have a + applied to your letter grade. If you do not get any “Exceeds Expectations” grades, then you will have a - applied. (Therefore, if you get exactly one (1) “Exceeds Expectations” grade, you will have neither a plus nor minus on your letter grade.)

If you do not meet the criteria for at least a D,- then you will get an F.

Course Policies

Students in this class are responsible for reading, understanding, and following all of the course policies listed here. By continuing to take this course, you indicate your agreement to follow all the policies. If there is any policy that you do not understand, please ask. Ignorance will not be

Assessment

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CS 115 Database Systems, fall 2022

Syllabus v1.2.1

Course Description

Fundamental concepts of database management systems. Topics include: data models (relational, object-oriented, and others); the SQL query language; implementation techniques of database management systems (storage and index structures, concurrency control, recovery, and query processing); management of unstructured and semi-structured data; and scientific data collections.

Learning Outcomes

Upon completion of the course, students will be able to:

1. Explain database design concepts (including ACID, CAP, eventual consistency, and vertical/horizontal scaling) for both relational

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2. Write queries that correctly and efficiently retrieve (SELECT), INSERT, UPDATE, and DELETE data in relational (SQL) databases, as well as queries that perform equivalent operations in several types of NoSQL databases.
3. Design a relational (SQL) database schema from a problem description and express the design as an Entity-Relationship (ER) or UML diagram or SQL language schema.
4. Enforce constraints on the data in relational (SQL) databases using schema constraints in the SQL language as well as using triggers/rules.
5. Prove that a relational (SQL) database is or is not in a particular normal form (3NF, BCNF/3.5NF, or 4NF), and transform a database schema into a particular normal form.
6. Choose the appropriate database system to use for a particular application, from relational (SQL) or several types of NoSQL databases.
7. Design databases for each of several types of NoSQL databases.
8. Use software design patterns that facilitate application code interfacing with database systems.

Prerequisites

- **CS 15 (Data Structures)** or *Graduate Student* standing
 - Grad students are expected to have had some sort of Data Structures course (e.g., CS 15, CS 205, or a Data Structures programming course from another college).
- Ability to program in some high-level language (e.g., C, C++, Java, etc.) and a willingness to learn new languages
 - You will receive explicit instruction on the SQL language in this

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Javascript (and possibly other languages), and you will be

expected to get up to speed on your own with what you need to know to follow along with the examples. However, you will not be expected to write complete programs yourself in any of these languages; just a few lines at the most.

- In order to best understand some of the example and database concepts, you should be familiar with basic object-oriented programming concepts (classes, methods, abstraction/interfaces, etc.) as well as common data structures, such as trees

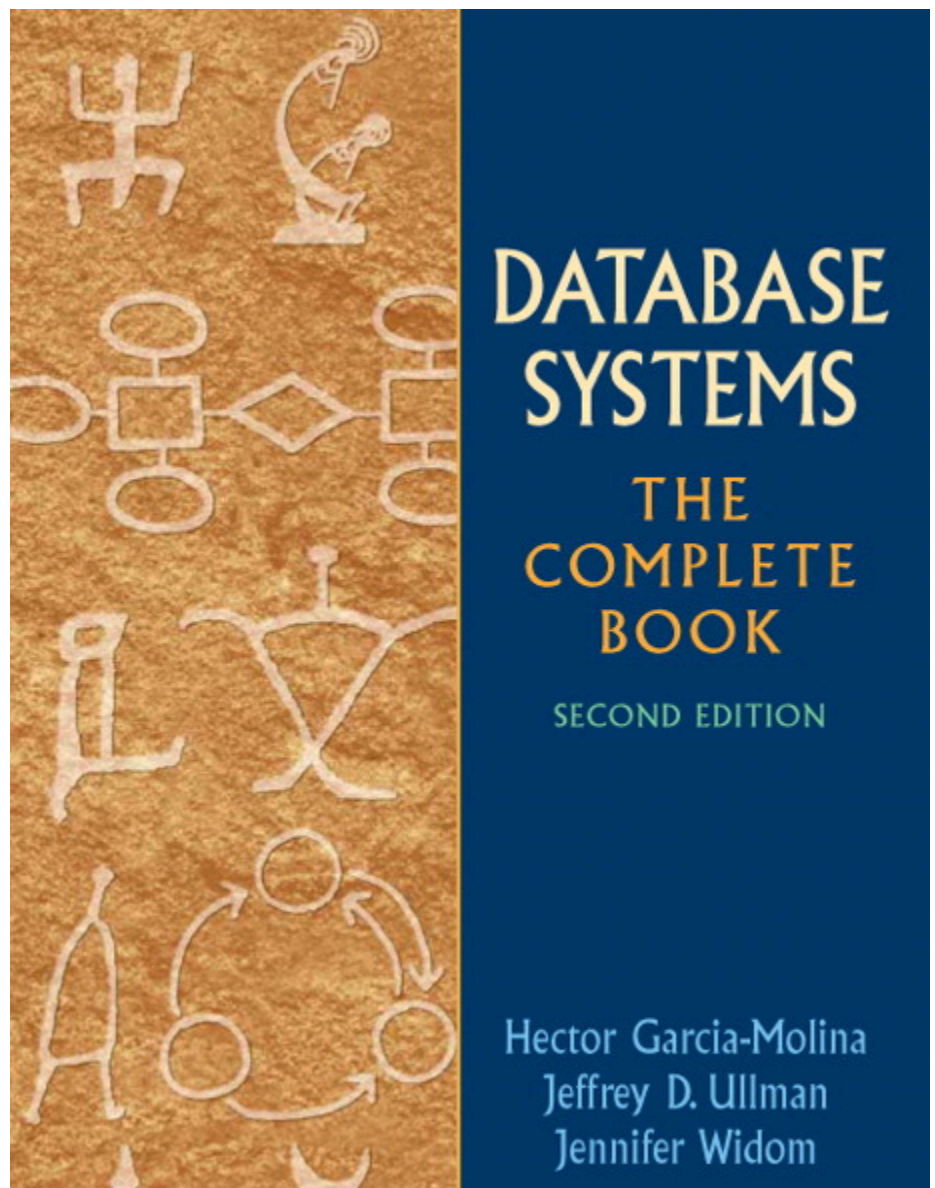
Course Materials

Required Textbook:

Database Systems: The Complete Book, 2nd Edition. H. Garcia-Molina, J. Ullman, and J. Widom (c) 2009. Pearson.

ISBN-13:
9780131873254

(Hardback, paperback, and digital editions are all fine. New used



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rented/borrowed are also fine. But it must be the 2nd edition.)

You will also need a laptop or desktop computer at home that you can either install database software on, or run a virtual machine that has the database software on it. Windows, Mac, and Linux systems are all acceptable, and all required software is freely available online.

Class Times & Location

This class will be held *in-person* for fall 2021. (Keep an eye on your email for announcements from the university or AS&E schools about COVID-19 precautions, and for announcements from the course instructor.)

- Tuesdays & Thursdays, 6:00 – 7:15 PM
- Joyce Cummings Center, room 160

The first class will be Tuesday, Sept. 6, 2022. The last day of class will be Thursday, Dec. 8, 2022. The final exam will be **Friday, Dec. 16, 2022, at 7:00 pm.**

Instructor & TAs

Dave Lillethun is the instructor. TAs are Tomislav Zabcic-Matic and Mona Ma.

Office hours and contact information will be posted in [Canvas](#).

Assessments

Take-Home Tests

There will be several tests (every other week, 6 in total) which you will

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tests. They will not be proctored and will not be strictly timed, but will be released at a certain time and due at a certain later time. You may work on the test any time before the deadline and take as long as you like, up to the deadline. Materials you are allowed to use when working on tests are:

- While taking the test, you may refer to the textbook, your notes (including previous tests, quizzes, class exercises, etc.), the Internet, and generally any other pre-written/recorded sources.
- You *may* **not** *collaborate or discuss* test questions or answers with anyone else (whether they are a student in this class or not). This also means that you may not post online (e.g., Stack Overflow, Reddit, etc.) to ask questions pertaining to the test; you may only use information online that already existed before you looked at the test.
- **You must write all answers in your own words.** Even though you may refer to other materials to help you figure out the answers, you may not plagiarize those materials on your answers. Even making minor modifications after copying is forbidden. You must *completely rewrite/rephrase it* **using your own words**.
- If ever in doubt, ask the instructor *before* you do something. It is better to ask for permission than forgiveness, in this case.

In-Class Quizzes

There will be several quizzes (every 2-3 weeks, 5 in total), which you will take in class. You will be given **30 minutes** to complete the quiz at the beginning of class on days when quizzes are given, and normal class will be held for the rest of the period following the test. **Quizzes are cumulative**, and you may be asked about any material covered in any lecture or required reading assignment from the textbook

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Quizzes will be proctored, and you *may* **not** use any calculators, other electronic devices, books, or notes **except** for a **one-page note sheet**. You will be allowed to prepare a single letter size (8.5"x11") sheet of paper with notes that you may use during the exam. You may write on both sides, and there are no rules about text size or margins, but it *must be* **hand written** (unless a disability accommodation allows otherwise). However, you will not be allowed to use any magnification devices (except for routine corrective eyewear, such as glasses), so you must write it such that you will be able to read it. While you are welcome to discuss what to put on the note sheet with other students, **each student must hand write their own copy of the note sheet**. *Photocopies, and other kinds of copies, will not be permitted.*

You will be required to turn in your note sheet with your quiz. Among other reasons, this will force you to rewrite a new one for each quiz, which is beneficial for several reasons:

- **The act of writing things down will help you to remember.**

Therefore, rewriting the things you need help remembering several times will help you remember them better than just writing them once. (Also, hand writing it will help you remember better than typing would.)

- There will be **new material on each quiz** that you'll want to add to your notes sheet.
- You should also remove old material from the notes sheet. If you find that you were able to remember something without the notes sheet, you may decide to remove it. More importantly, **you'll want to make room for the new material** that you're adding.

Final Exam

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The final exam will take place at **7:00 pm on Friday, Dec. 16** in the usual classroom. It will have the same rules, format, and types of questions as the quizzes (including the notes sheet), with the same types and formats of questions, except that it will be fully cumulative. All material from the entire class will be eligible for exam questions. The length of the exam will be approximately twice that of a quiz. (So in essence, it's a longer, more fully cumulative quiz.)

Late Work Policy

In this class, each student starts the semester with **5 “late tokens”** that are each worth an extra day (24 hours) on a take home test assignment. (With the exception that you will get Saturday and Sunday as a bundle for 1 token. Since tests will be due on Fridays, that means 1 token gives you until Sunday, 2 tokens until Monday, and an additional token for every day past then that you need.) There are no penalties for lateness as long as you use a sufficient number of late tokens. If you will not have work completed by the deadline, then **email the instructor** (contact information is in Canvas) **before the deadline** and tell him how many extra days you need. A corresponding number of late tokens will be deducted. You don't need to provide any excuses, just the new date when you will submit the work. As long as you send that email before the assignment deadline, your late submission will be graded for full credit. (However, if there is an emergency situation that makes it impossible or unreasonable to contact us before the deadline, then just send the email as soon after the deadline as is reasonable.)

Note that the quizzes and final exam are scheduled to occur in specific class periods, and therefore the deadline cannot be extended. However,

make-up quizzes and exams can be provided if you miss it due to a

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miss any of these.

Course Grades

Each test, quiz, or exam will *not* be given a percentage score or letter grade, but rather each individual question will be rated as “Exceeds Expectations”, “Meets Expectations”, “Needs Improvement”, or “Not Assessable” (“N/A” for short). These mean the following:

- **Exceeds Expectations** – This assessment shows that you have significant proficiency in the relevant topic, beyond the minimum expected by the course learning expectations.
- **Meets Expectations** – This assessment shows that you have some proficiency in the relevant topic, in line with the course learning expectations.
- **Needs Improvement** – This assessment shows some knowledge but does not yet demonstrate full proficiency in the topic.
- **Not Assessable** – This assessment is too incomplete to accurately assess whether you have met expectations or not, or it shows little evidence of knowledge or skills. This could be, for example, a question in which not all parts were attempted fully, was left blank, or was partially answered but too much was missing to really be able to assess the student’s knowledge/skills.

Here is what you need to do to earn each letter grade. You must meet *all* of the criteria for a letter grade in order to get that grade.

- A
 - Meets Expectations (or higher) on a total of **at least 90.0%** of test questions, and
 - Exceeds Expectations on **at least 30.0%** of test questions, and

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- Exceeds Expectations on **at least 30.0%** of quiz & exam questions
- B
 - Meets Expectations (or higher) on a total of **at least 70.0%** of test questions, and
 - Exceeds Expectations on **at least 10.0%** of test questions, and
 - Meets Expectations (or higher) on a total of **at least 70.0%** of quiz & exam questions, and
 - Exceeds Expectations on **at least 10.0%** of quiz & exam questions
- C
 - Meets Expectations (or higher) on a total of **at least 55.0%** of test questions, and
 - Meets Expectations (or higher) on a total of **at least 55.0%** of quiz & exam questions
- D
 - Meets Expectations (or higher) on a total of **at least 50.0%** of test questions, and
 - Meets Expectations (or higher) on a total of **at least 50.0%** of quiz & exam questions

Students who meet all the criteria for a letter grade, **plus at least 10.0%** (percentage points) more than required Exceed Expectations on *each* of the test questions category and the quiz & exam questions category, **will receive a plus (+)** on their grade. (e.g., B+ for 30% of test questions and 40% of quiz & exam question, or C+ for 10% of test questions and 10% of quiz & exam questions Exceed Expectations.)

Students who are **within 5.0%** (percentage points) of the required number of questions Meeting or Exceeding Expectations on each of the test

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Expectations on *each* of the test questions category and the quiz & exam questions category, **will receive that letter with a minus (-)**, rather than the next lower letter. (e.g., Exceeding Expectations on 10% of test questions and 20% of quiz questions, while Meeting (or Exceeding) Expectations on 80% of test questions and 80% of quiz & exam questions is a B-. However, if you instead Exceed Expectations on only 10% of quiz questions, then that's a C+, and if you Meet (or Exceed) Expectations on only 79% of test questions, then that's a C no matter how many questions you Exceeded Expectations on.)

Students who do not meet the criteria for at least a D- will receive an F.

Course Policies

Students in this class are responsible for reading, understanding, and following all of the [course policies listed here](#). For fall 2021, there are also additional [COVID-19 policies](#) for the course. By continuing to take this course, you indicate your agreement to follow all the policies. If there is any policy that you do not understand, please ask. Ignorance will not be accepted as an excuse for violating any policies.

Changes

This syllabus and any policies for this course are subject to change during the semester in response to changing conditions. Such changes are at the sole discretion of the course instructor. If any changes are made, the appropriate documents will be updated and the change will be announced to the entire class in a timely fashion.

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COMP 150-SEN

Software Engineering Foundations, Spring 2019

Staff			
Name	Office	E-mail	Office Hours (also available by appointment)
Jeff Foster	211 Halligan	jfoster@cs.tufts.edu	MW 1:15-2:00pm
Tomoki Shibata	Wed: Halligan extension 007; Fri: Halligan 209	tshibata@cs.tufts.edu	W 4:15-5:15pm, F 9:30-10:30am

Information	
Location	Bromfield-Pearson Room 002
Time	MW 3:00-4:15
Midterm	March 13, in class
Final	May 3, 3:30-5:30pm
Textbooks	There are no required textbooks

Description

The scale of modern software systems is truly amazing. We regularly use software that is hundreds-of-thousands to millions of lines of code, and that software, while certainly not bug-free, mostly does what it is supposed to. Achieving this kind of scale has been the result of steady progress in *software engineering* over the last several decades.

In this class, we will study the foundations of software engineering, focusing on the core principles and ideas that enable us to build large-scale software systems. Our focus will be on the coding side of software engineering. Equally important, but deferred to a different course, is the people/team side of software engineering. We will explore ideas such as abstraction, modularity, architecture, specification, testing, and debugging, among others.

The course will be conducted in Java, and will include a short introduction to Java at the beginning. During the course, students will complete a number of programming assignments, and the course will also include some reading assignments.

Prerequisites: COMP 40, graduate standing, or instructor consent. If you are not sure whether you meet the prerequisites for the course, please contact the instructor.

Schedule

Week	Monday	Wednesday
Jan 14	(No class)	No class
Jan 21	MLK Day No class	Introduction Java Reading Assignment #1
Jan 28	Java Project 1	Java LinkedList.java List.java

Feb 4	Java ArrayList.java	Java
Feb 11	Design Patterns Reading Assignment #2	Design Patterns Project 2
Feb 18	<i>President's Day</i> No class	(Feb 20) Modularity Discussion Design Patterns (Feb 21) Design Patterns ooc.c Reflection
Feb 25	Software Architecture Reading Assignment #3	Testing
Mar 4	Fuzz Testing Discussion Testing	Testing
Mar 11	Midterm Review	Midterm
Mar 18	<i>Spring Recess</i> No class	<i>Spring Recess</i> No class
Mar 25	Refactoring Project 3	Program Verification
Apr 1	<i>Guest Lecture:</i> Milod Kazerounian, Program Verification (cont'd) .	<i>Guest Lecture:</i> Diogenes Nunez , Garbage Collection Reading Assignment #4
Apr 8	Security java-sec-example.tar.gz	No Silver Bullet discussion Project 4
Apr 15	<i>Patriots' Day</i> No class	Debugging How Failures Come to Be The Scientific Method Reproducing Problems
Apr 22	Simplifying Problems Fixing the Defect Reading Assignment #5	ACM Code of Ethics
Apr 29	What Makes a Great Software Engineer Final Exam Review	(No class)

Reading Assignments

- (Due Jan 30) Tony Hoare, [Null References: The Billion Dollar Mistake](#)
 - It's okay to read the show notes instead of watch the video.
 - Q1: What was the most interesting thing you learned from the presentation?
 - Q2: Do you think null pointers really were a billion dollar mistake?
- (Due Feb 20) D.L. Parnas, [On the Criteria To Be Used in Decomposing Systems into Modules](#)
 - Q1: Briefly describe one important lesson about modularity that the paper describes and that you think is still relevant today.
 - Q2: Computing has advanced significantly since this paper was written. Briefly describe one challenge (not necessarily with modularity) or perspective that, while realistic in 1972, does not apply today.
- (Due Mar 4) Miller, Fredriksen, and So, [An Empirical Study of the Reliability of UNIX Utilities](#)
 - Q1: Briefly describe one strength and one weakness of fuzz testing.
 - Q2: Two modern fuzz testers are [american fuzzy lop](#) and [libfuzzer](#). Pick one, look through the web page(s) for it, and briefly describe one way it improves on the original fuzz testing work of Miller et al.
- (Due Apr 10) Brooks, [No Silver Bullet: Essence and Accidents of Software Engineering](#)
 - Q1: Briefly discuss one contrast Brooks draws between Software Engineering and other areas or science or engineering.
 - Q2: Brooks discusses several "Hopes for the silver", but his comments are about the state of the art in 1987. Describe one "hope for the silver" that you see today, but that Brooks did not see back then. Or, if you are a pessimist, explain why there is no hope.
- (Due Apr 29) Li, Ko, and Zhu, [What makes a great software engineer?](#)
 - Q1: What was the most surprising characteristic to you of a great software engineering, as described by the paper?
 - Q2: What are some things you might want to do on your own, after this course ends, to become a great software engineer?

Projects

- [Project 1](#) - Java ADTs and the Adapter Pattern
- [Project 2](#) - Design Patterns in Java
- [Project 3](#) - A Unit Testing Framework
- [Project 4](#) - Java on Rails

Exams

- [Sample questions](#) for the midterm ([solutions](#))
- [Midterm sample solutions](#)
- [Final sample solutions](#)

Syllabus

Syllabus subject to change until the start of the semester

Prerequisites

COMP 40, graduate standing, or instructor consent. If you are not sure whether you meet the prerequisites for the course, please contact the instructor.

List of Topics (Tentative)

Below is an approximate list of topics for class. The exact topics will be determined based on the pace of the class.

- Java programming, including classes, objects, inheritance, interfaces, delegation, Java generics, and the Java runtime
- Abstract data types, modularity, information hiding
- Design patterns, including for concurrency
- Software architecture
- Program specification and verification
- Object-oriented refactoring
- Testing
- Debugging
- Program synthesis
- Special topics (TBA)

Office Hours and Web Forum

Office hours for the instructional staff will be posted on the course web page a few days into the semester.

While we will provide assistance with assignments during office hours, you are responsible for developing and debugging your own programs. Do not rely on the instructional staff to make your project work.

Important announcements will be made in class or on Piazza. Please make it a habit to check Piazza daily, and/or sign up to receive email when updates are posted to Piazza. You may also use the class web forum to ask general questions of interest to the class as a whole, e.g., administrative issues or project clarification questions. Please do not post any information that would violate the University's Academic Integrity Policy.

Grading

You are responsible for all material discussed in class and posted on the class web page, including announcements, deadlines, policies, etc. Your final course grade will be determined according to the following percentages:

Projects/homework	50%
Readings	9%
Midterm	20%
Final	20%
Meet your professor	1%

Any request for reconsideration of any grading on coursework **must** be submitted within **one week** of when it is returned. Exam regrading requests must be made in writing. Any coursework submitted for reconsideration may be regraded in its entirety, which could result in a lower score if warranted.

Final course grades will be curved as necessary, based on each student's total numeric score for all coursework at the end of the semester. **Important:** Completing the programming assignments is an essential part of the course. Therefore, **we may fail any student who does not make a good-faith attempt on all course projects**, regardless of the student's performance or scores on the other coursework.

Programming Projects

Projects must be submitted electronically following the instructions given in class. Projects **may not** be submitted by any other means (e.g., please do not email your projects to us). It is **your responsibility** to test your program and verify that it works properly before submitting. All projects are due at 11:59pm on the day indicated on the project assignment, according to the submission server's internal clock.

Projects may be submitted up to 24 hours late for a 10% penalty. For example, a project that would earn 90 points for an on-time submission will earn 81 (which is 90 times 0.90) if submitted late. Note that your project score as it appears on the project submission server will not include any late penalties. Any penalties will be incorporated into the final project grade on the grade server.

Exam Scheduling

The class includes a midterm and a final exam. **Tentative** dates for the exams will be posted on the class web site. The exact dates will be confirmed later. We will let you know the exact dates well in advance.

Academic Resources

The [Academic Resource Center](#) offers a range of services for students.

Accommodations for Students with Disabilities

Tufts University values the diversity of our students, staff, and faculty, recognizing the important contribution each student makes to our unique community. Tufts is committed to providing equal access and support to all qualified students through the provision of reasonable accommodations so that each student may fully participate in the Tufts experience. If you have a disability that requires reasonable accommodations, please contact the [Student Accessibility Services](#) office to make an appointment with an SAS representative to determine appropriate accommodations. Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for their provision.

Please also contact the instructor to discuss any necessary accommodations.

Excused Absences

You are expected to attend class regularly, complete course assignments on time, and take exams at the scheduled times. If you are unable to fulfill these requirements due to absence for a good reason, the instructor will excuse the absence and provide accommodation. Events that justify an excused absence include:

- Religious observances
- Mandatory military obligation
- Illness of the student or illness of an immediate family member
- Participation in university activities at the request of university authorities
- Compelling circumstances beyond the student's control (e.g., death in the family, required court appearance, etc.)

It is **your responsibility** to inform the instructor **in advance** of intended religious observances. Notice must be provided **immediately** upon an exam date being announced or confirmed for an absence to be excused.

The policies for excused absences **do not** apply to project assignments. Projects will be assigned with sufficient time to allow students to carry out the work even with other responsibilities. In cases of extremely serious documented illness of lengthy duration or other protracted, severe emergency situations, the instructor may consider extensions depending on the specific circumstances.

Absences stemming from job interviews, traffic or transportation problems, personal travel, and similar will not be excused.

Academic Integrity

The university's [Academic Integrity Policy](#) will be strictly enforced.

Unless otherwise specified, programming projects are to be written **individually**. Therefore, cooperation or use of unauthorized materials on projects is a violation of the Academic Integrity Policy. **Project solutions may not be posted online**. Any evidence of this, or of unacceptable use of computer accounts, use of unauthorized materials or cooperation on exams or quizzes, or other possible academic integrity violations will be reported.

For learning the course concepts, students are welcome to study together or to receive help from anyone else. You may discuss with others the project requirements, the features of the programming languages used, what was discussed in class and in the class web forum, and general syntax errors. Examples of questions that would be allowed are "Does a Java class definition end in a semicolon?" or "What does a 'class not found' error indicate?", because they convey no information about the contents of a project.

When it comes to actually writing a project assignment, other than help from the instructional staff a project must solely and entirely be your own work. Working with another student or individual, or using anyone else's work in any way except as noted in this paragraph, is a violation of the Academic Integrity Policy. You may not discuss design of any part of a project with anyone except the instructor or teaching assistants. Examples of questions you may not ask others might be "How did you implement this part of the project?" or "Please look at my code and help me find my stupid syntax error!". You may not use any disallowed source of information in creating either their project design or code. When writing projects you are free to use ideas or short fragments of code from published textbooks or publicly available information, but the specific source must be cited in a comment in the relevant section of the program.

Violations of the Code of Academic Integrity may include, but are not limited to:

1. Failing to do all or any of the work on a project by yourself, other than assistance from the instructional staff.
2. Using any ideas or any part of another person's project, or copying any other individual's work in any way.
3. Giving any parts or ideas from your project, including test data, to another student.
4. Allowing any other students access to your program on any computer system.
5. Transferring any part of a project to or from another student or individual by any means, electronic or otherwise.

If you have any question about a particular situation or source then consult with the instructor in advance. Should you have difficulty with a programming assignment you should **see the instructional staff in office hours**, and not solicit help from anyone else in violation of these rules.

Right to Change Information

Although every effort has been made to be complete and accurate, unforeseen circumstances arising during the semester could require the adjustment of any material given here. Consequently, given due notice to students, the instructor reserves the right to change any information on this syllabus or in other course materials.

Resources

- Java Basics
 - [Java Development Kit \(JDK\) Download](#)
 - [JDK 11 Documentation](#)
 - [JDK 11 API](#)
- Java Textbooks (read via [Safari Books Online](#))
 - [Head First Java](#)
 - [The Java Programming Language](#)
 - [Effective Java](#)
 - [Java Cookbook](#)
- Related Papers
 - Garlan and Shaw, [An Introduction to Software Architecture](#)
 - Klees et al., [Evaluating Fuzz Testing](#)
- Other
 - [Explaining Code using ASCII Art](#)
 - [Why Programs Fail \(Slides\)](#)